

9

Appendix: Case Studies

As mentioned in Chapter 1, the presentation in this book is by and large applicable to any rigid, fixed-wing airplane. Throughout this book we have used numerical data for the F-18 HARV — a combat aircraft — to numerically illustrate the various concepts discussed. Examples of other aircraft, including missiles, parafoils, and airships, have appeared either in the chapter-end Exercise Problems or as Examples within the text. We have also commented in Chapter 7 on the matter of scale and how the developments in this book apply equally well to UAV's. In this Appendix, we provide supplementary material to add to the contents of the first eight chapters; however, it is not necessary to read or be acquainted with any of the contents of this Appendix to appreciate the earlier presentation. We shall take up two case studies in this Appendix: First, that of a light General Aviation (GA) airplane, and secondly, an example of a rigid airship.

9.1. Example of GA airplane

We shall use this case study to present two different aspects of flight dynamics calculations that we could not present in the previous chapters.

1. Estimation of aerodynamic characteristics (the aero coefficients and derivatives) of an airplane — In the examples presented previously, the aerodynamic data (called “aero data” for short) was presented in the form of tables or figures or curve-fitted formulas. In general, aero data can be obtained by one of three means: by analytical/semi-empirical methods, by the use of numerical methods (CFD), or from wind tunnel tests. Often, during an aircraft design and development process, all three methods are employed at various points of time, and the data obtained from each of the methods may be blended to create a final database. In this section, we shall first take up an example GA aircraft and demonstrate the use of the analytical/ semi-empirical method to estimate the aero data.

2. Modal (longitudinal and lateral-directional) stability analysis using the first-order form of the equations — In Chapters 5 and 6, we have presented the small-perturbation equations for the dynamics of the airplane in longitudinal flight and in lateral-directional flight, respectively, in second-order form. The second-order form is physically meaningful and convenient to derive the literal approximations to the various modes (short period, Dutch roll, etc.), as we have already seen. On the other hand, the first-order form of the linearized equations for the modal dynamics is commonly used to numerically evaluate the eigenvalues of the ‘A’ matrix and hence determine the stability of the modes. In the interest of completeness, we present here the small-perturbation equations for the longitudinal and lateral-directional dynamics rewritten in first-order form. Subsequently, we use the aero data estimated for the GA aircraft to populate the ‘A’ matrices for the longitudinal and lateral-directional dynamics and evaluate the stability of the respective modes



Figure 9.1: A six-seater business transport, Piper PA-32-300 Cherokee (Photo Courtesy: Arpingstone).

<https://upload.wikimedia.org/wikipedia/commons/f/f2/Piper.pa32.cherokee.six.g-bxwp.arp.jpg>

9.1.1 Aero data estimation

The GA aircraft model used here is a six-seater business transport similar to the Piper Cherokee shown in Figure 9.1. The aerodynamic estimation procedure follows along the lines as illustrated in Ref.~1. The aircraft is a low-wing conventional configuration with a single engine (Textron Lycoming piston type of 360hp with 3-blade propeller) mounted in a tractor arrangement. A trapezoidal wing planform of taper ratio 0.5 has been used. Wing airfoil section varies along span from NACA23018 at root to NACA23012 at wing tip, however, wing aerodynamic data for the aircraft has been estimated by assuming constant airfoil section NACA23018 throughout the span. NACA0012 airfoil section has been used for both horizontal and vertical tails. Relevant aircraft parameters are provided in Tables 1,2 and 3.

TABLE 9.1

GA aircraft geometry, mass and inertia data

Acceleration due to gravity	$g = 9.81m/s^2$	Mass of aircraft	$m = 1859.73kg$
Reference area (wing planform area)	$S_w = 16.35m^2$	Moments of inertia (about principal axes?)	$I_{xx} = 3355.65kg - m^2$
Wing root chord (NACA 23018)	$c_r = 2.02m$		$I_{yy} = 4180.57kg - m^2$
Wing tip chord (NACA 23012)	$c_t = 1.01m$		$I_{zz} = 6140.66kg - m^2$
Wing span	$b_w = 10.75m$	Wing incidence setting angle	$i_w = 0deg$
Wing dihedral angle	$\Gamma = 5deg$	Wing zero-lift angle of attack	$\alpha_{L=0} = -1.1deg;$
Wing quarter-chord sweep angle	$\Lambda_{c/2} = 5.38deg$	CG location from aircraft nose	$X_{CG} = 2.39m$
Fuselage length	$l_f = 7.89m$	Horizontal tail planform area	$S_{HT} = 3.45m^2$
Fuselage side projected area	$S_f = 8.12m^2$	Horizontal tail aspect ratio	$AR_{HT} = b_{HT}^2/S_{HT} = 4.0$
Maximum fuselage width diameter	$D_f = 1.37m$	Horizontal tail incidence setting angle	$i_t = 0deg$
Vertical tail planform area	$S_{VT} = 2.16m^2$	HT aerodynamic center location (from aircraft nose)	$X_{AC}^{HT} = 7.62m$
Vertical tail aspect ratio	$AR_{VT} = b_{VT}^2/S_{VT} = 1.5$	VT aerodynamic center location (from aircraft nose)	$X_{AC}^{VT} = 7.28m$
Aileron surface area	$S_{\delta a} = 0.82m^2$	Elevator surface area	$S_{\delta e} = 1.38m^2$

Aileron span	$l_{\delta a} = 2.15m$	Rudder surface area	$S_{\delta r} = 0.43m^2$
--------------	------------------------	---------------------	--------------------------

TABLE 9.2

GA aircraft flight data

Density of air at flight altitude	$\rho_{air} = 1.22kg/m^3$	Speed of sound at flight altitude	$a = 340m/s$
Dynamic pressure ratio at the tail	$\eta = 0.95$		

TABLE 9.3

GA aircraft engine data

Maximum engine power	$P_{max} = 268452N - m/s$	Number of propeller blades	$N = 3$
Propeller diameter	$D = 1.99m$	Propeller disk distance from aircraft CG	$l_{prop} = 2.39m$

Calculated values:

Wing mean aerodynamic chord: $\bar{c} = \frac{2}{3} c_r \left(\frac{1+\lambda+\lambda^2}{1+\lambda} \right) = \frac{2}{3} \times 2.02 \times \left(\frac{1+0.5+0.5^2}{1+0.5} \right) = 1.57m;$

Wing aspect ratio: $AR_w = b_w^2/S_w = \frac{10.75^2}{16.35} = 7.07;$

Wing taper ratio: $\lambda = c_t/c_r = 0.5;$

Horizontal tail span: $b_{HT} = \sqrt{4.0 \times 3.45} = 3.71m;$

Vertical tail span: $b_{VT} = \sqrt{1.5 \times 2.16} = 1.8m;$

Vertical tail AC height above FRL: $z_{VT} = \frac{4}{9} \times b_{VT} = \frac{4}{9} \times 1.8 = 0.8m;$

Propeller disc area: $S_{prop} = \frac{\pi D^2}{4} = \frac{\pi \times 1.99^2}{4} = 3.11m^2;$

CG location, SM, Tail arms:

HT moment arm length: $l_{HT} = X_{AC}^{HT} - X_{CG} = 7.62 - 2.39 = 5.23m;$

VT moment arm length: $l_{VT} = X_{AC}^{VT} - X_{CG} = 7.28 - 2.39 = 4.89m;$

HT volume ratio: $V_{HT} = \frac{S_{HT} \times l_{HT}}{S_w \times \bar{c}} = \frac{3.45 \times 5.23}{16.35 \times 1.57} = 0.7 ;$

$$\text{VT volume ratio: } V_{VT} = \frac{S_{VT} \times l_{VT}}{S_w \times b_w} = \frac{2.16 \times 4.89}{16.35 \times 10.75} = 0.06;$$

Lift-curve slope:

Wing airfoil lift curve slope, $c_{l\alpha,w} = 0.1 / \text{deg} = 5.73 / \text{rad}$ (From Ref. 1 for NACA23018);

HT lift curve slope, $c_{l\alpha,HT} = 0.109 / \text{deg} = 6.24 / \text{rad}$;

VT lift curve slope, $c_{l\alpha,VT} = 0.109 / \text{deg} = 6.24 / \text{rad}$;

For subsonic speeds, the lift curve slope of straight tapered wings can be determined using the expressions:

$$\text{Wing lift-curve slope: } C_{L\alpha,w} = \frac{2\pi AR_w}{2 + \sqrt{\frac{AR_w^2 \beta^2}{k^2} \left(1 + \frac{\tan^2 \Lambda_{c/2}}{\beta^2}\right) + 4}}; \beta = \sqrt{1 - Ma^2}, k = C_{l\alpha,w} / 2\pi, \Lambda_{c/2} \text{ is}$$

the sweep angle of mid-chord of the wing. At low subsonic speed ($\beta \approx 1$) and $\left(\frac{\tan^2 \Lambda_{c/2}}{\beta^2}\right) \ll 1$, the above expression simplifies to

$$C_{L\alpha,w} = \frac{c_{l\alpha,w}}{1 + (c_{l\alpha,w} / \pi AR_w)}.$$

$$\text{Wing lift curve slope: } C_{L\alpha,w} = \frac{c_{l\alpha,w}}{1 + (c_{l\alpha,w} / \pi AR_w)} = \frac{5.73}{1 + (5.73 / 3.14 \times 7.07)} = 4.55 / \text{rad};$$

$$\text{HT life curve slope: } C_{L\alpha,HT} = \frac{c_{l\alpha,HT}}{1 + (c_{l\alpha,HT} / \pi AR_{HT})} = \frac{6.24}{1 + (6.24 / 3.14 \times 4.0)} = 4.17 / \text{rad};$$

$$\text{VT life curve slope: } C_{L\alpha,VT} = \frac{c_{l\alpha,VT}}{1 + (c_{l\alpha,VT} / \pi AR_{VT})} = \frac{6.24}{1 + (6.24 / 3.14 \times 1.5)} = 2.68 / \text{rad};$$

Wing lift coefficient at zero angle-of-attack:

$$C_{L0,w} = C_{L\alpha,w} \times \alpha_{L=0} \times \pi / 180 = 4.55 \times 1.1 \times \pi / 180 = 0.087;$$

$$\text{Non-dimensionalized CG location from airplane nose: } h_{CG} = X_{CG} / \bar{c} = \frac{2.39}{1.57} = 1.52$$

Assuming static margin: $SM = 15\%$

$$h_{NP} - h_{CG} = 0.15;$$

$$h_{NP} = 1.52 + 0.15 = 1.67; X_{NP} = h_{NP} \times \bar{c} = 1.67 \times 1.57 = 2.62m$$

$$h_{AC}^{wb} = h_{NP} - V_{HT} \left\{ \frac{C_{L\alpha,HT}}{C_{L\alpha,w}} \right\} \left(1 - \frac{d\varepsilon}{d\alpha} \right)$$

$$\begin{aligned} \text{Downwash angle: } \varepsilon &= \frac{2C_{L,w}}{\pi AR_w} = \underbrace{\frac{2C_{L0,w}}{\pi AR_w}}_{\varepsilon_0} + \underbrace{\frac{2C_{L\alpha,w}}{\pi AR_w}}_{\varepsilon_\alpha} (\alpha + i_w) \\ &= \frac{2 \times 0.087}{\frac{\pi \times 7.07}{0.008}} + \frac{2 \times 4.55}{\frac{\pi \times 7.07}{0.409}} \times \alpha; \end{aligned}$$

$$h_{AC}^{wb} = h_{NP} - V_{HT} \left\{ \frac{C_{L\alpha,HT}}{C_{L\alpha,w}} \right\} \left(1 - \frac{d\varepsilon}{d\alpha} \right) = 1.67 - 0.7 \times \frac{4.17}{4.55} \times (1 - 0.409) = 1.29;$$

$$\text{Wing aerodynamic center location from aircraft nose: } X_{AC}^{wb} = h_{AC}^{wb} \times \bar{c} = 1.29 \times 1.57 = 2.02m;$$

$$\text{Oswald efficiency factor: } e = 0.9; K = \frac{1}{\pi AR_w e} = \frac{1}{\pi \times 7.07 \times 0.9} = 0.05;$$

$$\text{Residual pitching moment coefficient at aerodynamic center: } C_{m,ac} = -0.005;$$

Pitching moment coefficient and trim calculation:

*Contributions to pitching moment coefficient from fuselage and propeller can be significant and must be modelled. Both these effects are neglected here and only contributions from wing and horizontal tail are considered.

$$C_{m\alpha,w} = \left(\frac{X_{CG} - X_{AC}^{wb}}{\bar{c}} \right) \times C_{L\alpha,w} = \left(\frac{2.39 - 2.02}{1.57} \right) \times 4.55 = 1.07/rad;$$

$$C_{m\alpha,HT} = -\eta' V_{HT} C_{L\alpha,HT} (1 - \varepsilon_\alpha) = -0.95 \times 0.7 \times 4.17 \times (1 - 0.409) = -1.64/rad;$$

$$C_{m\alpha} = C_{m\alpha,w} + C_{m\alpha,HT} = 1.07 - 1.64 = -0.57;$$

$$\begin{aligned} C_{m0} &= C_{mAC,w} + C_{m0,HT} \\ &= -0.005 + \eta' V_{HT} C_{L\alpha,HT} (-i_t + \varepsilon_0) \\ &= -0.005 + 0.95 \times 0.7 \times 4.17 \times \left(0 \times \frac{\pi}{180} + 0.008 \right) = 0.017 \end{aligned}$$

$$\text{Trim angle of attack: } \alpha^* = -\frac{C_{m0}}{C_{m\alpha}} = -\frac{0.017}{-0.57} = 0.03rad = 1.7deg \text{ (For zero elevator deflection)}$$

Airplane trim lift coefficient:

$$\begin{aligned} C_L^* &= C_{L\alpha,w} \times (\alpha^* - \alpha_{L=0}) + \eta' \frac{S_{HT}}{S_w} C_{L\alpha,HT} \left(\alpha^* - \frac{d\varepsilon}{d\alpha} \alpha^* - \varepsilon_0 + i_t \right) \\ &= 4.55 \times (1.7 - (-1.1)) \times \frac{\pi}{180} + 0.95 \times \frac{3.45}{16.35} \times 4.17 \\ &\quad \times \left\{ 1.7 \times \frac{\pi}{180} \times (1 - 0.409) - 0.008 - 0 \times \frac{\pi}{180} \right\} = 0.22 + 0.008 = 0.23; \end{aligned}$$

Trim speed:

$$V^* = \sqrt{\frac{2 W}{\rho S_w C_L^*}} = \sqrt{\frac{2}{1.22} \times \frac{1859.73 \times 9.81}{16.35} \times \frac{1}{0.23}} = 89.18m/s;$$

$$C_{L0} = C_{L\alpha,w} (0 - \alpha_{L=0}) + \eta' \frac{S_{HT}}{S_w} C_{L\alpha,HT} (i_t - \varepsilon_0)$$

$$= 4.55 \times 1.1 \times \frac{\pi}{180} + 0.95 \times \frac{3.45}{16.35} \times 4.17 \times (-0 - 0.008) = 0.08$$

$$C_{L\alpha} = C_{L\alpha,w} + \eta' \frac{S_{HT}}{S_w} C_{L\alpha,HT} \left(1 - \frac{d\varepsilon}{d\alpha}\right) = 4.55 + 0.95 \times \frac{3.45}{16.35} \times 4.17 \times (1 - 0.409) = 5.04/rad;$$

$$C_{mq1} = -2\eta' V_{HT} C_{L\alpha,HT} \times (l_{HT}/\bar{c}) = -2 \times 0.95 \times 0.7 \times 4.17 \times \frac{5.23}{1.57} = -18.47/rad$$

$$C_{m\dot{\alpha}} = 0.0/rad; C_{L\dot{\alpha}} = 0.0/rad;$$

$$C_{Lq1} = 2\eta' V_{HT} C_{L\alpha,HT} = 2 \times 0.95 \times 0.7 \times 4.17 = 5.56/rad;$$

$$C_{Dq1} = 2KC_L^* C_{Lq1} = 2 \times \frac{1}{\pi \times 0.9 \times 7.07} \times 0.23 \times 5.56 = 0.13/rad;$$

$$C_D^* = C_{D0} + KC_L^{*2} = 0.017 + \frac{0.23^2}{\pi \times 0.9 \times 7.07} = 0.02;$$

$$\text{Thrust required: } T_R = \frac{1}{2} \rho V^{*2} S C_D^* = 0.5 \times 1.22 \times 89.18^2 \times 16.35 \times 0.02 = 1586.4N;$$

$$\text{Power required: } P_R = T_R V^* = 1586.4 \times 89.18 = 141475.09J = 189.6hp;$$

$$C_{D\alpha}^* = 2KC_L^* C_{L\alpha} = \frac{2 \times 0.23 \times 5.04}{\pi \times 0.9 \times 7.07} = 0.12/rad$$

$$\bar{q}^* = \frac{1}{2} \rho V^{*2} = \frac{1}{2} \times 1.22 \times 89.18^2 = 4851.37N/m^2$$

$$\left(\frac{\bar{q}^* S c}{I_{yy}}\right) = \frac{4851.37 \times 16.35 \times 1.57}{4180.57} = 29.78/s^2$$

$$\left(\frac{g}{V^*}\right) = \frac{9.81}{89.18} = 0.11; \frac{\bar{q}^* S}{W} = \frac{4851.37 \times 16.35}{1859.73 \times 9.81} = 4.35; Ma^* = \frac{V^*}{a} = \frac{89.18}{340.0} = 0.26$$

$$\left(\frac{g}{V^*}\right) \left(\frac{\bar{q}^* S}{W}\right) = 0.11 \times 4.35 = 0.48; \frac{\bar{c}}{2V^*} = \frac{1.57}{2 \times 89.18} = 0.009;$$

$$C_{LMa} = C_L^* \frac{Ma^*}{1 - Ma^{*2}} = 0.23 \times \frac{0.26}{1 - 0.26^2} = 0.064; C_{DMA} = 0.0$$

Longitudinal mode calculations:

Short period mode:

From Eqs. 2.33 and 2.34:

$$\begin{aligned} (\omega_n^2)_{SP} &= -\left(\frac{\bar{q}^* S c}{I_{yy}}\right) C_{m\alpha} \\ &= -29.78 \times -0.57 = 16.97 \end{aligned}$$

$$\omega_{nSP} = 4.12 \text{ rad/s}; T_{SP} = \frac{2\pi}{\omega_{SP}} = \frac{2 \times 3.14}{4.12} = 1.52 \text{ s}$$

$$\begin{aligned} (2\zeta\omega_n)_{SP} &= -\left(\frac{\bar{q}^* S c}{I_{yy}}\right) \left(\frac{c}{2V^*}\right) (C_{mq1} + C_{m\dot{\alpha}}) \\ &= -29.78 \times \frac{1.57}{2 \times 89.18} \times (-18.47 + 0) = 4.84; \\ \zeta_{SP} &= \frac{4.84}{2\omega_{nSP}} = \frac{4.84}{2 \times 4.12} = 0.59 \end{aligned}$$

Phugoid mode:

From Eqs. 5.35 and 5.36:

$$\begin{aligned} (\omega_n^2)_p &= \left(\frac{g}{V^*}\right)^2 \left[\frac{\bar{q}^* S}{W} (Ma^* C_{LMa} + 2C_L^*) \right] \\ &= 0.11^2 \times [4.35 \times (0.26 \times 0.064 + 2 \times 0.23)] = 0.025 \\ \omega_{nPh} &= \sqrt{0.025} = 0.16 \text{ rad/s}; T_{Ph} = \frac{2\pi}{\omega_{nPh}} = \frac{2 \times 3.14}{0.16} = 39.66 \text{ s} \\ (2\zeta\omega_n)_{Ph} &= \left(\frac{g}{V^*}\right) \left[\frac{\bar{q}^* S}{W} (Ma^* C_{DMa} + 2C_D^*) \right] \approx \left(\frac{g}{V^*}\right) \left[\frac{\bar{q}^* S}{W} 2C_D^* \right] \\ &= 0.11 \times 4.35 \times 2 \times 0.02 = 0.02; \\ \zeta_{Ph} &= \frac{0.02}{2 \times 0.16} = 0.06; \end{aligned}$$

9.1.2 First-order form of the small-perturbation longitudinal dynamics equations

Beginning with the longitudinal small-perturbation dynamics equations in level flight presented in Chapter 5, we now recast them in first-order form. The state variables of interest are: ΔV , $\Delta \gamma$, $\Delta \alpha$, and $\Delta \dot{\alpha}$.

The velocity dynamics at the slow phugoid time scale comes from Eq. 5.42:

$$\begin{aligned} \frac{\Delta \dot{V}}{V^*} &= \left(\frac{g}{V^*}\right) \left\{ -\frac{\bar{q}^* S}{W} \left(\Delta C_D + 2C_D^* \frac{\Delta V}{V^*} \right) - \Delta \gamma \right\} \\ &= \left(\frac{g}{V^*}\right) \left\{ -\frac{\bar{q}^* S}{W} \left(Ma^* C_{DMa} \frac{\Delta V}{V^*} + C_{D\alpha} \Delta \alpha + C_{Dq1} \Delta \dot{\alpha} \left(\frac{c}{2V^*} \right) + 2C_D^* \frac{\Delta V}{V^*} \right) - \Delta \gamma \right\} \\ &= \left(\frac{g}{V^*}\right) \left\{ -\frac{\bar{q}^* S}{W} \left(\{Ma^* C_{DMa} + 2C_D^*\} \frac{\Delta V}{V^*} + C_{D\alpha} \Delta \alpha + C_{Dq1} \Delta \dot{\alpha} \left(\frac{c}{2V^*} \right) \right) - \Delta \gamma \right\} \end{aligned}$$

$$\begin{aligned}
&= -\left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\{Ma^*C_{DMa} + 2C_D^*\}\frac{\Delta V}{V^*} - \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{D\alpha}\Delta\alpha \\
&\quad - \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{Dq1}\left(\frac{c}{2V^*}\right)\Delta\dot{\alpha} - \left(\frac{g}{V^*}\right)\Delta\gamma
\end{aligned} \tag{9.1}$$

From Eq. 5.43:

$$\begin{aligned}
\Delta\dot{\gamma} &= \left(\frac{g}{V^*}\right)\left\{\frac{\bar{q}^*S}{W}\left(\Delta C_L + 2C_L^*\frac{\Delta V}{V^*}\right)\right\} \\
&= \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\left\{\left(\{Ma^*C_{LMa} + 2C_L^*\}\frac{\Delta V}{V^*} + C_{L\alpha}\Delta\alpha + C_{Lq1}\Delta\dot{\alpha}\left(\frac{c}{2V^*}\right)\right)\right\} \\
&= \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\{Ma^*C_{LMa} + 2C_L^*\}\frac{\Delta V}{V^*} + \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{L\alpha}\Delta\alpha + \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{Lq1}\left(\frac{c}{2V^*}\right)\Delta\dot{\alpha}
\end{aligned} \tag{9.2}$$

$$\begin{aligned}
\frac{\Delta\dot{V}}{V^*} &= -\left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\{Ma^*C_{DMa} + 2C_D^*\}\frac{\Delta\dot{V}}{V^*} - \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{D\alpha}\Delta\dot{\alpha} - \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{Dq1}\left(\frac{c}{2V^*}\right)\Delta\ddot{\alpha} \\
&\quad - \left(\frac{g}{V^*}\right)\Delta\dot{\gamma} \\
&= -\left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\{Ma^*C_{DMa} + 2C_D^*\}\frac{\Delta\dot{V}}{V^*} - \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{D\alpha}\Delta\dot{\alpha} - \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{Dq1}\left(\frac{c}{2V^*}\right)\Delta\ddot{\alpha} \\
&\quad - \left(\frac{g}{V^*}\right)\left\{\left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\{Ma^*C_{LMa} + 2C_L^*\}\frac{\Delta V}{V^*} + \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{L\alpha}\Delta\alpha + \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{Lq1}\left(\frac{c}{2V^*}\right)\Delta\dot{\alpha}\right\}
\end{aligned} \tag{9.3}$$

The pitch dynamics at the fast short period time scale is obtained from Eq. 5.44:

$$\Delta\ddot{\theta} = \Delta\ddot{\alpha} + \Delta\ddot{\gamma} = \left(\frac{\bar{q}^*Sc}{I_{yy}}\right)\Delta C_m \tag{9.4}$$

$$\begin{aligned}
\Delta\ddot{\alpha} &= \left(\frac{\bar{q}^*Sc}{I_{yy}}\right)\Delta C_m - \Delta\ddot{\gamma} \\
&= \left(\frac{\bar{q}^*Sc}{I_{yy}}\right)\left\{Ma^*C_{mMa}\frac{\Delta V}{V^*} + C_{m\alpha}\Delta\alpha + C_{mq1}\Delta\dot{\alpha}\left(\frac{c}{2V^*}\right)\right\} \\
&\quad - \left\{\left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\left(\{Ma^*C_{LMa} + 2C_L^*\}\frac{\Delta\dot{V}}{V^*} + C_{L\alpha}\Delta\dot{\alpha} + C_{Lq1}\Delta\ddot{\alpha}\left(\frac{c}{2V^*}\right)\right)\right\}
\end{aligned}$$

$$\Delta\ddot{\alpha}\left\{1 + \left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)C_{Lq1}\left(\frac{c}{2V^*}\right)\right\} = -\left(\frac{g}{V^*}\right)\left(\frac{\bar{q}^*S}{W}\right)\{Ma^*C_{LMa} + 2C_L^*\}\frac{\Delta\dot{V}}{V^*}$$

$$\begin{aligned}
& + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) M a^* C_{mMa} \frac{\Delta V}{V^*} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) C_{m\alpha} \Delta \alpha \\
& + \left\{ \left(\frac{\bar{q}^* S c}{I_{yy}} \right) \left(\frac{c}{2V^*} \right) C_{mq1} - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{L\alpha} \right\} \Delta \dot{\alpha} \\
\Delta \ddot{\alpha} & \left\{ 1 + \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{Lq1} \left(\frac{c}{2V^*} \right) \right\} \\
& = \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{ M a^* C_{LMa} + 2C_L^* \} \left[\begin{array}{c} \{ M a^* C_{DMa} + 2C_D^* \} \frac{\Delta V}{V^*} \\ + C_{D\alpha} \Delta \alpha \\ + C_{Dq1} \left(\frac{c}{2V^*} \right) \Delta \dot{\alpha} + \left\{ \left(\frac{g}{V^*} \right) / \left(\frac{\bar{q}^* S}{W} \right) \right\} \Delta \gamma \end{array} \right] \\
& + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) M a^* C_{mMa} \frac{\Delta V}{V^*} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) C_{m\alpha} \Delta \alpha \\
& + \left\{ \left(\frac{\bar{q}^* S c}{I_{yy}} \right) \left(\frac{c}{2V^*} \right) C_{mq1} - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{L\alpha} \right\} \Delta \dot{\alpha} \\
& = \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right) \{ M a^* C_{LMa} + 2C_L^* \} \Delta \gamma + \\
& + \left\{ \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{ M a^* C_{LMa} + 2C_L^* \} C_{D\alpha} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) C_{m\alpha} \right\} \Delta \alpha \\
& + \left\{ \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{ M a^* C_{LMa} + 2C_L^* \} \{ M a^* C_{DMa} + 2C_D^* \} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) M a^* C_{mMa} \right\} \frac{\Delta V}{V^*} \\
& + \left\{ \left(\frac{\bar{q}^* S c}{I_{yy}} \right) \left(\frac{c}{2V^*} \right) C_{mq1} - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{L\alpha} \right\} \\
& + \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 C_{Dq1} \left(\frac{c}{2V^*} \right) \{ M a^* C_{LMa} + 2C_L^* \} \Delta \dot{\alpha}
\end{aligned}$$

Let us define

$$\begin{aligned}
k_{\ddot{\alpha}} &= \left\{ 1 + \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{Lq1} \left(\frac{c}{2V^*} \right) \right\} \\
k_{\gamma} &= \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right) \{ M a^* C_{LMa} + 2C_L^* \} \\
k_{\alpha} &= \left\{ \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{ M a^* C_{LMa} + 2C_L^* \} C_{D\alpha} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) C_{m\alpha} \right\}
\end{aligned}$$

$$k_{\dot{\alpha}} = \left\{ \left(\frac{\bar{q}^* S c}{I_{yy}} \right) \left(\frac{c}{2V^*} \right) C_{mq1} - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{L\alpha} \right\} + \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 C_{Dq1} \left(\frac{c}{2V^*} \right) \{Ma^* C_{LMa} + 2C_L^*\}$$

$$k_V = \left\{ \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{Ma^* C_{LMa} + 2C_L^*\} \{Ma^* C_{DMA} + 2C_D^*\} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) Ma^* C_{mMa} \right\}$$

The above equations can be re-cast as

$$k_{\ddot{\alpha}} \Delta \ddot{\alpha} = k_V (\Delta V / V^*) + k_\gamma \Delta \gamma + k_\alpha \Delta \alpha + k_{\dot{\alpha}} \Delta \dot{\alpha} \quad (9.5)$$

Introducing the equality relation

$$\Delta \dot{\alpha} = \Delta \dot{\alpha} \quad (9.6)$$

and collecting term by term from Eqs. 9.1-9.6, matrix form of the longitudinal perturbation dynamics can be written as

$$\begin{bmatrix} \Delta \dot{V} / V^* \\ \Delta \dot{\gamma} \\ \Delta \dot{\alpha} \\ \Delta \ddot{\alpha} \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}}_{Along} \begin{bmatrix} \Delta V / V^* \\ \Delta \gamma \\ \Delta \alpha \\ \Delta \dot{\alpha} \end{bmatrix} \quad (9.7)$$

Evaluating the matrix at the trim condition:

$$a_{11} = - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) \{Ma^* C_{DMA} + 2C_D^*\} = -0.11 \times 4.35 \times (0.26 \times 0 + 2 \times 0.02) = -0.019;$$

$$a_{12} = - \left(\frac{g}{V^*} \right) = -0.11; a_{13} = - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{D\alpha} = -0.11 \times 4.35 \times 0.12 = -0.058;$$

$$a_{14} = - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{Dq1} \left(\frac{c}{2V^*} \right) = -0.11 \times 4.35 \times 0.13 \times 0.009 = 0.0006;$$

$$a_{21} = \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) \{Ma^* C_{LMa} + 2C_L^*\} = 0.11 \times 4.35 \times (0.26 \times 0.064 + 2 \times 0.23) = 0.23;$$

$$a_{22} = 0.0; a_{23} = \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{L\alpha} = 0.11 \times 4.35 \times 5.04 = 2.41;$$

$$a_{24} = \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{Lq1} \left(\frac{c}{2V^*} \right) = 0.11 \times 4.35 \times 5.56 \times 0.009 = 0.024;$$

$$a_{31} = 0; a_{32} = 0; a_{33} = 0; a_{34} = 1.0;$$

$$k_V = \left(\left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{Ma^* C_{LMa} + 2C_L^*\} \{Ma^* C_{DMA} + 2C_D^*\} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) Ma^* C_{mMa} \right)$$

$$= 0.11^2 \times 4.35^2 \times (0.26 \times 0.064 + 2 \times 0.23) \times (0 + 2 \times 0.02) + 29.78 \times 0.26 \times 0 = 0.004;$$

$$k_{\ddot{\alpha}} = \left\{ 1 + \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{Lq1} \left(\frac{c}{2V^*} \right) \right\} = 1 + 0.11 \times 4.35 \times 5.56 \times 0.009 = 1.024;$$

$$k_{\gamma} = \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right) \{Ma^* C_{LMa} + 2C_L^*\} = 0.11^2 \times 4.35 \times (0.26 \times 0.064 + 2 \times 0.23) = 0.025;$$

$$k_{\alpha} = \left(\left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 \{Ma^* C_{LMa} + 2C_L^*\} C_{D\alpha} + \left(\frac{\bar{q}^* S c}{I_{yy}} \right) C_{m\alpha} \right)$$

$$= 0.11^2 \times 4.35^2 \times (0.26 \times 0.064 + 2 \times 0.23) \times 0.12 + 29.78 \times -0.57 = -16.96;$$

$$k_{\dot{\alpha}} = \left\{ \left(\frac{\bar{q}^* S c}{I_{yy}} \right) \left(\frac{c}{2V^*} \right) C_{mq1} - \left(\frac{g}{V^*} \right) \left(\frac{\bar{q}^* S}{W} \right) C_{L\alpha} \right\} + \left(\frac{g}{V^*} \right)^2 \left(\frac{\bar{q}^* S}{W} \right)^2 C_{Dq1} \left(\frac{c}{2V^*} \right) \{Ma^* C_{LMa} + 2C_L^*\}$$

$$= (29.78 \times 0.009 \times -18.47) - (0.11 \times 4.35 \times 5.04) + (0.11^2 \times 4.35^2 \times 0.13 \times 0.009) \times (0.26 \times 0.064 + 2 \times 0.23) = -4.95 - 2.41 + 0.0001 = -7.36$$

$$a_{41} = k_{\gamma}/k_{\ddot{\alpha}} = \frac{0.005}{1.024} = 0.0049; a_{42} = k_{\gamma}/k_{\ddot{\alpha}} = \frac{0.025}{1.024} = 0.024;$$

$$a_{43} = k_{\alpha}/k_{\ddot{\alpha}} = -\frac{16.96}{1.024} = -16.56; a_{44} = k_{\dot{\alpha}}/k_{\ddot{\alpha}} = -\frac{7.36}{1.024} = -7.19;$$

$$A_{long} = \begin{bmatrix} -0.019 & -0.11 & -0.058 & 0.0 \\ 0.23 & 0.0 & 2.41 & 0.024 \\ 0.0 & 0.0 & 0.0 & 1.0 \\ 0.005 & 0.024 & -16.96 & -7.19 \end{bmatrix}$$

Computed eigenvalues of A_{long} are:

$$\lambda_{SP} = -3.59 \pm j2.01 \text{ which gives } \omega_{nSP} = 4.12 \text{ rad/s (4.12); } \zeta_{SP} = 0.87 (0.59)$$

$$\lambda_{Ph} = -0.0078 \pm j0.1589 \text{ which gives } \omega_{nPh} = 0.16 \text{ rad/s (0.16); } \zeta_{Ph} = 0.05 (0.06)$$

Note: In bracket are the values calculated earlier from approximations. While other parameters match well, significant difference in Short period damping can be observed. This difference is primarily due to the additional term containing $C_{L\alpha}$ appearing in A_{long} .

9.1.3. Lateral-Directional Aerodynamics Parameters

(**Note:** Contributions from fuselage, propeller and interference effects are not modeled in the following. On occasions, their contribution may be significant and may need to be considered.)

Airplane side-force coefficients:

Dihedral effect: $C_{Y\beta,w} = -0.0001\Gamma = -0.0001 \times 5 \times \frac{\pi}{180} \approx 0$;

Vertical location of wing root quarter-chord above CG line, $z_w = 0.305m$;

Sidewash, σ :

$$\left(1 + \frac{d\sigma}{d\beta}\right)\eta'_{VT} = 0.724 + 3.06 \frac{S_{VT}/S_w}{1 + \cos\Lambda_c/4} + \frac{0.4z_w}{D_{f,max}} + 0.009AR_w$$
$$= 0.724 + 3.06 \times \frac{2.16/16.35}{2} + \frac{0.4 \times 0.305}{1.3716} + 0.009 \times 7.07 = 1.08;$$

Vertical tail contribution:

$$C_{Y\beta,VT} = -0.95 \times C_{L\alpha,VT} \times \frac{S_{VT}}{S_w} \times \left(1 + \frac{d\sigma}{d\beta}\right)\eta'_{VT} = -0.95 \times 2.68 \times \frac{2.16}{16.35} \times 1.08 = -0.36;$$

Total: $C_{Y\beta} = C_{Y\beta,w} + C_{Y\beta,VT} = -0.36/rad$;

Airplane yawing moment coefficients:

Dihedral effect: $C_{n\beta,w} = -0.075 \times \Gamma \times C_L = -0.075 \times 5 \times \frac{\pi}{180} \times 0.23 = -0.001$;

Vertical tail contribution: $C_{n\beta,VT} = -C_{Y\beta,VT} \times \frac{l_{VT}}{b_w} = -(-0.29) \times \frac{4.88}{10.75} = 0.13$;

Total: $C_{n\beta} = C_{n\beta,w} + C_{n\beta,VT} = -0.001 + 0.13 \approx 0.13$;

Airplane rolling moment coefficients:

$$z = z_{VT} \cos\alpha - l_{VT} \sin\alpha = 0.8 \times \cos(1.7) - 4.873 \times \sin(1.7) = 0.66;$$

Wing contribution: $C_{l\beta,w} = -\frac{\Gamma C_{L\alpha,w} c_r b_w}{6S_w} = -5 \times \frac{\pi}{180} \times 4.55 \times 2.02 \times \frac{10.75}{6 \times 16.35} = -0.09$;

Vertical tail contribution: $C_{l\beta,VT} = (z/b_w)C_{Y\beta,VT} = \frac{0.66}{10.75} \times -0.36 = -0.02$;

$$C_{l\beta} = C_{l\beta,w} + C_{l\beta,VT}$$
$$= -0.09 - 0.02 = -0.11/rad;$$

Roll rate derivatives,

$$C_{Yp2,VT} = \frac{2}{b_w} \times (z - z_{VT}) \times C_{Y\beta,VT} = \frac{2}{10.75} \times (0.66 - 0.8) \times -0.36 = 0.01;$$

$$C_{Yp2,w} = 0; C_{Yp2,\Gamma} = 0;$$

$$C_{Yp2} = C_{Yp2,VT} + C_{Yp2,w} + C_{Yp2,\Gamma} = 0.01;$$

$$C_{lp2,VT} = 2 \times \left(\frac{z}{b_w}\right) \times \left(\frac{z-z_{VT}}{b_w}\right) \times C_{Y\beta,VT} = 2 \times \frac{0.66}{10.75} \times \frac{0.66-0.8}{10.75} \times -0.36 = -0.0006;$$

$$C_{lp2,w} = -\frac{1}{6}(C_{L\alpha} + C_D) = -\frac{1}{6} \times (5.04 + 0.02) = -0.84;$$

$$C_{lp2} = C_{lp2,VT} + C_{lp2,w} = -0.84;$$

$$C_{np2,w} = -\frac{1}{6}(C_L - C_{D\alpha}) = -\frac{1}{6} \times (0.23 - 0.12) = -0.02;$$

$$\begin{aligned} C_{np2,VT} &= -\frac{2}{b_w} \times (l_{VT}\cos\alpha + z_{VT}\sin\alpha) \times \left(\frac{z-z_{VT}}{b_w}\right) \times C_{Y\beta,VT} \\ &= -\frac{2}{10.75} \times (4.873 \times \cos(1.7) + 0.8 \times \sin(1.7)) \times \frac{0.66-0.8}{10.75} \times -0.36 = -0.004; \end{aligned}$$

$$C_{np2} = C_{np2,w} + C_{np2,VT} = -0.02/rad;$$

Yaw rate derivatives:

$$\begin{aligned} C_{Yr1,VT} &= -\frac{2}{b_w} \times (l_{VT}\cos\alpha + z_{VT}\sin\alpha) \times C_{Y\beta,VT} \\ &= -\frac{2}{10.75} \times (4.873 \times \cos(1.7) + 0.8 \times \sin(1.7)) \times -0.36 = 0.33; \end{aligned}$$

$$C_{lr1,w} = C_L/3 = 0.23/3 = 0.08;$$

$$\begin{aligned} C_{lr1,VT} &= -\frac{2}{b_w^2} \times (l_{VT}\cos\alpha + z_{VT}\sin\alpha) \times (z_{VT}\cos\alpha - l_{VT}\sin\alpha) \times C_{Y\beta,VT} \\ &= -\frac{2}{10.75^2} \times (4.873 \cos(1.7) + 0.8 \sin(1.7)) \times (0.8 \cos(1.7) - 4.873 \sin(1.7)) \times -0.36 \\ &= 0.02; \end{aligned}$$

$$C_{lr1} = C_{lr1,w} + C_{lr1,VT} = 0.1;$$

$$\begin{aligned} C_{nr1,VT} &= \frac{2}{b^2} \times (l_{VT}\cos\alpha + z_{VT}\sin\alpha)^2 \times C_{Y\beta,VT} \\ &= \frac{2}{10.75^2} \times (4.873 \times \cos(1.7) + 0.8 \times \sin(1.7))^2 \times -0.36 = -0.15; \end{aligned}$$

$$C_{nr1,w} = 0;$$

$$C_{nr1} = C_{nr1,w} + C_{nr1,VT} = -0.15;$$

Lateral-directional mode calculations:

$$\bar{q}^* = \frac{1}{2} \rho V^{*2} = \frac{1}{2} \times 1.22 \times 89.18^2 = 4851.37 N/m^2$$

$$\left(\frac{\bar{q}^* S b}{I_{xx}} \right) = \frac{4851.37 \times 16.35 \times 10.75}{3355.65} = 254.11/s^2$$

$$\left(\frac{\bar{q}^* S b}{I_{zz}} \right) = \frac{4851.37 \times 16.35 \times 10.75}{6140.66} = 138.86/s^2$$

$$\left(\frac{\bar{q}^* S}{m} \right) = \frac{4851.37 \times 16.35}{1859.73} = 42.65 m/s^2$$

$$C_{lp2} = -0.84; \frac{b}{2V^*} = \frac{10.75}{2 \times 89.18} = 0.06; \frac{g}{V^*} = \frac{9.81}{89.18} = 0.11$$

$$C_{n\beta} = 0.13; C_{Y\beta} = -0.36; C_{nr1} = -0.15; C_{l\beta} = -0.11; C_{lr1} = 0.1$$

$$C_{nr2} = -\frac{C_D}{3} = -\frac{0.02}{3} = -0.01$$

$$C_{lr2} = \frac{C_L}{3} = \frac{0.23}{3} = 0.08$$

$N_\beta = \left(\frac{\bar{q}^* S b}{I_{zz}} \right) C_{n\beta}$ $= 138.86 \times 0.13 = 18.05/s^2$	$N_{r2} = \left(\frac{\bar{q}^* S b}{I_{zz}} \right) C_{nr2} \left(\frac{b}{2V^*} \right)$ $= 138.86 \times -0.01 \times 0.06 = -0.08/s^2$
$Y_\beta = \left(\frac{\bar{q}^* S}{m} \right) C_{Y\beta}$ $= 42.65 \times -0.36 = -15.35 m/s^2$	$\mathcal{L}_\beta = \left(\frac{\bar{q}^* S b}{I_{xx}} \right) C_{l\beta}$ $= 254.11 \times -0.11 = -27.95/s^2$
$\mathcal{L}_{p2} = \left(\frac{\bar{q}^* S b}{I_{xx}} \right) C_{lp2} \left(\frac{b}{2V^*} \right)$ $= 254.11 \times -0.84 \times 0.06$ $= -12.81/s^2$	$N_{r1} = \left(\frac{\bar{q}^* S b}{I_{zz}} \right) C_{nr1} \left(\frac{b}{2V^*} \right)$ $= 138.86 \times -0.15 \times 0.06 = -1.25/s^2$
$\mathcal{L}_{r1} = \left(\frac{\bar{q}^* S b}{I_{xx}} \right) C_{lr1} \left(\frac{b}{2V^*} \right)$ $= 254.11 \times 0.1 \times 0.06 = 1.52/s^2$	$\mathcal{L}_{r2} = \left(\frac{\bar{q}^* S b}{I_{xx}} \right) C_{lr2} \left(\frac{b}{2V^*} \right)$ $= 254.11 \times 0.08 \times 0.06 = 1.22/s^2$

Roll mode: $\lambda_r = \mathcal{L}_{p2} = -12.81/s^2$

Dutch roll mode:

$$\begin{aligned}\omega_{nDR}^2 &= N_\beta + \left(\frac{g}{V^*}\right) \left[Y_\beta N_{r2} + \left(\frac{\mathcal{L}_\beta}{\mathcal{L}_{p2}}\right) \right] \\ &= 18.05 + \left(\frac{9.81}{89.18}\right) \left[-15.35 \times -0.08 + \left(\frac{-27.95}{-12.81}\right) \right] = 18.42;\end{aligned}$$

$$\omega_{nDR} = 4.29 \text{ rad/s};$$

$$\begin{aligned}2\zeta_{DR}\omega_{nDR} &= -N_{r1} - \left(\frac{g}{V^*}\right) \left[Y_\beta + \left(\frac{\mathcal{L}_{r1}}{\mathcal{L}_{p2}}\right) \right] \\ &= 1.25 - \left(\frac{9.81}{89.18}\right) \left[-15.35 + \left(\frac{1.52}{-12.81}\right) \right] = 2.95;\end{aligned}$$

$$\zeta_{DR} = \frac{2.95}{2 \times 4.29} = 0.34;$$

Spiral mode:

$$\begin{aligned}\lambda_S &= \left(\frac{g}{V^*}\right) \frac{[\mathcal{L}_\beta N_{r2} - N_\beta \mathcal{L}_{r2}]}{\lambda_r \omega_{nDR}^2} \\ &= \left(\frac{9.81}{89.18}\right) \frac{((-27.95 \times -0.08) - (18.05 \times 1.22))}{-12.81 \times 18.42} = 0.009/s;\end{aligned}$$

9.1.4. Lateral-directional perturbation dynamics model

From Table 6.1,

$$\Delta \ddot{\mu} = -\mathcal{L}_{r1} \Delta \dot{\beta} + \left[\mathcal{L}_\beta + \left(\frac{g}{V^*}\right) Y_\beta \mathcal{L}_{r2} \right] \Delta \beta + \mathcal{L}_{p2} \Delta \dot{\mu} + \left(\frac{g}{V^*}\right) \mathcal{L}_{r2} \Delta \mu \quad (9.8)$$

$$\Delta \ddot{\beta} = -\left[-N_{r1} - \left(\frac{g}{V^*}\right) Y_\beta \right] \Delta \dot{\beta} - \left[N_\beta + \left(\frac{g}{V^*}\right) Y_\beta N_{r2} \right] \Delta \beta + \left(\frac{g}{V^*}\right) \Delta \dot{\mu} - \left(\frac{g}{V^*}\right) N_{r2} \Delta \mu \quad (9.9)$$

By introducing equality relations,

$$\Delta \dot{\mu} = \Delta \dot{\mu}, \quad \Delta \dot{\beta} = \Delta \dot{\beta} \quad (9.10)$$

One may arrive at the matrix form of the above equations put together in variables $[\Delta \mu \quad \Delta \dot{\mu} \quad \Delta \beta \quad \Delta \dot{\beta}]$ as,

$$\begin{bmatrix} \Delta \dot{\mu} \\ \Delta \ddot{\mu} \\ \Delta \dot{\beta} \\ \Delta \ddot{\beta} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ (g/V^*)\mathcal{L}_{r2} & \mathcal{L}_{p2} & \{\mathcal{L}_\beta + (g/V^*)Y_\beta \mathcal{L}_{r2}\} & -\mathcal{L}_{r1} \\ 0 & 0 & 0 & 1 \\ -(g/V^*)N_{r2} & (g/V^*) & \{-N_\beta - (g/V^*)Y_\beta N_{r2}\} & \{N_{r1} + (g/V^*)Y_\beta\} \end{bmatrix}}_{A_{lat-dir}} \begin{bmatrix} \Delta \mu \\ \Delta \dot{\mu} \\ \Delta \beta \\ \Delta \dot{\beta} \end{bmatrix} \quad (9.11)$$

At our trim condition:

$$A_{lat-dir} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0.13 & -12.81 & -30.0 & -1.52 \\ 0 & 0 & 0 & 1 \\ 0.008 & 0.11 & -18.185 & -2.94 \end{bmatrix}$$

Computed values of $A_{lat-dir}$ are:

$$\lambda_S = 0.009; \lambda_r = -12.83;$$

$$\lambda_{DR} = -1.471 \pm j4.033, \text{ which gives } \omega_{nDR} = 4.29 \text{ rad/s}, \zeta_{DR} = 0.34$$

Control Effectiveness Parameter (From Chart in Fig. 7.9):

$$\text{Elevator control effectiveness parameter, } \tau_{\delta e} = \frac{\Delta \alpha_t}{\Delta \delta e} = 0.5;$$

$$\text{Rudder control effectiveness parameter, } \tau_{\delta r} = \frac{\Delta \alpha_{VT}}{\Delta \delta r} = 0.5;$$

$$\text{Aileron control effectiveness parameter, } \tau_{\delta a} = \frac{\Delta \alpha}{\Delta \delta a} = 0.22;$$

Control derivatives (calculated at trim $\alpha = 1.7 \text{ deg}$):

$$C_{L,\delta e} = \eta' \times \frac{S_{HT}}{S_w} \times C_{L\alpha,HT} \times \tau_{\delta e} = 0.95 \times \frac{3.45}{16.35} \times 4.17 \times 0.5 = 0.42/\text{rad};$$

$$C_{m,\delta e} = -\eta' V_{HT} C_{L\alpha,HT} \tau_{\delta e} = -0.95 \times 0.7 \times 4.17 \times 0.5 = -1.38/\text{rad};$$

$$C_{l\delta a} = -2 \left(\frac{C_{l\alpha,w} \tau_{\delta a}}{Sb} \right) \int_{0.5b}^{0.9b} c(y) y dy \quad (\text{for aileron running between 50-90\% of wing span})$$

$$= -2 \left(\frac{C_{l\alpha,w} \tau_{\delta a}}{Sb} \right) c_r \int_{0.5b}^{0.9b} \left[1 + \frac{2y}{b} (\lambda - 1) \right] y dy$$

$$= -2 \left(\frac{C_{l\alpha,w} \tau_{\delta a}}{Sb} \right) c_r \left[\frac{y^2}{2} + \frac{2y^3}{3b} (\lambda - 1) \right]_{0.5b}^{0.9b};$$

$$= -2 \times \left(\frac{5.73 \times 0.22}{16.35 \times 10.75} \right) \times 2.02 \times \left[\frac{0.9^2 - 0.5^2}{2} - \frac{2(0.9^3 - 0.5^3)}{3} \times (0.5 - 1) \right] \times 10.75^2$$

$$= -0.26 / \text{rad};$$

$$C_{n\delta a} \approx - \left(\frac{C_{D\alpha}}{C_{l\alpha,w}} \right) C_{l\delta a} = - \frac{0.12}{5.73} \times 0.26 = -0.01 / \text{rad};$$

$$C_{Y\delta a} = 0;$$

$$C_{n\delta r} = -V_{VT} \times \tau_{\delta r} \times C_{L\alpha_{VT}} = -0.06 \times 0.5 \times 2.68 = -0.08 \text{ /rad};$$

$$C_{Y\delta r} = (S_{VT}/S_w) \times \tau_{\delta r} \times C_{L\alpha_{VT}} = (2.16/16.35) \times 0.5 \times 2.68 = 0.18 \text{ /rad};$$

$$C_{l\delta r} = (S_{VT}/S_w) \times \tau_{\delta r} \times C_{L\alpha_{VT}} \times (z/b_w) = (2.16/16.35) \times 0.5 \times 2.68 \times \frac{0.66}{10.75} = 0.011 \text{ /rad}$$

9.2. Airship Dynamics

Airships, also known as blimps or dirigibles, are nothing but Helium (inert) or Hydrogen (flammable) filled balloons powered by engine/s producing thrust. Flexible (inflated by pressure equalization on outer and internal sides), semi-rigid (with some stiffening provided by inside structure) and rigid (with rigid structure inside) airships are buoyancy lifted aerial vehicles floating at altitudes at which weight of airship is neutralized by the buoyancy force. Usual shape of an airship is a doubly symmetric body of revolution as shown in Figure 9.2. Three points of interest on an airship are: Center of control volume (CV), which is the geometric center of the hull and Center of buoyancy (CB) where the buoyancy force acts; both are, in general, different from its Center of gravity (CG). Important features of an airship are the control surfaces similar to a rigid airplane with an engine located at the bottom. The control surfaces, rudder and elevons, provide the control in yaw and pitch/roll, respectively, and engine propellers may be thrust vectored for additional control. Further, ballonets (air-filled bags) fitted inside airship may also be used as a control device via discharge of air.

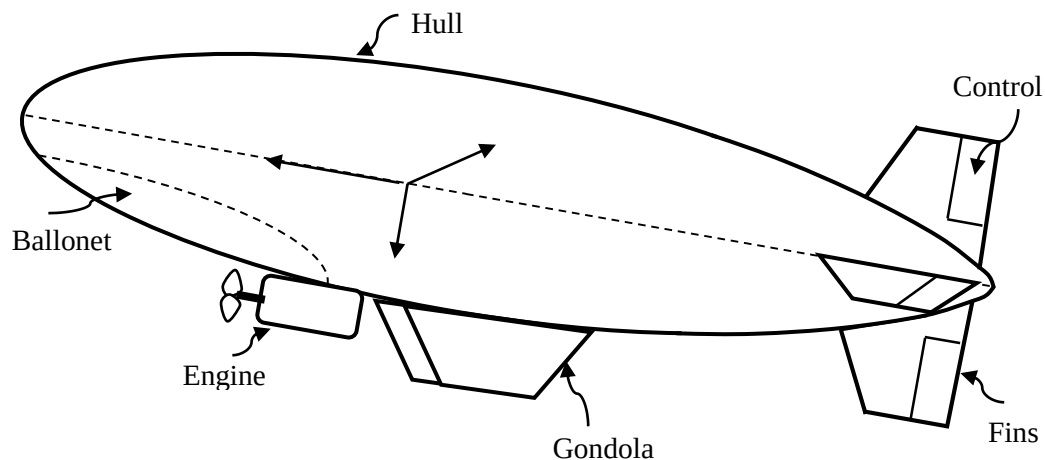


Figure 9.2: Airship nomenclature.

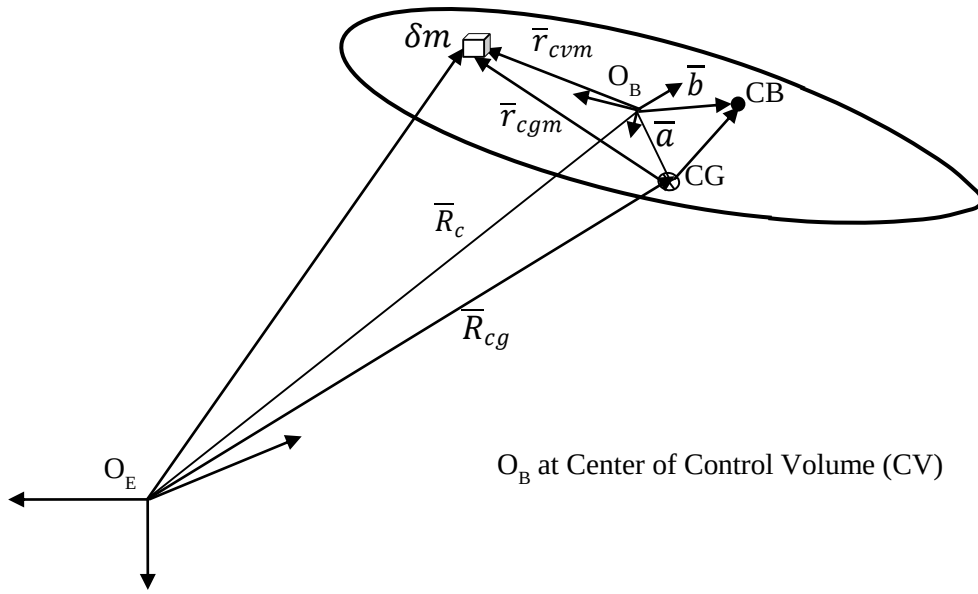


Figure 9.3: Body fixed axis system at the center of control volume.

9.2.1 Airship Equations of Motion

Six degrees of freedom motion of a rigid airship with fixed mass and assuming no aeroelastic effects can be described by three angular rotations and three translations motions similar to airplanes. Airship equations of motion are slightly different from that of airplanes mainly due to different choice of origin of body fixed system, which is conveniently taken to be the center of volume (CV) of the airship and not the center of gravity, which for the reasons explained later is supposed to lie closer to the gondola. As pictured in Fig. 9.3, the CG is located at distance $\bar{a} = [a_x \ a_y \ a_z]^T$ from the CV. The location of CB from the CV is $\bar{b} = [b_x \ b_y \ b_z]^T$. Angular velocity vector $\bar{\omega} = [p \ q \ r]^T$ and translation velocity vector $\bar{V} = [u \ v \ w]^T$ are defined in the body fixed axis system. Following the material presented in Chapter 8 (and a good part left as exercise!), now we write down the equations of airship motion.

Force equations (please refer to the original equations in Chapter 8 or wherever for comparison):

$$\bar{F} = m \frac{d\bar{V}}{dt} \Big|_B + m(\bar{\omega} \times \bar{V}) + m \frac{d(\bar{\omega} \times \bar{a})}{dt} \Big|_B + m\{\bar{\omega} \times (\bar{\omega} \times \bar{a})\} \quad (9.12)$$

The additional terms in Eq. 9.12 appear due to the choice of origin of the body-fixed axis which is not at CG but at the CV of the hull. Accounting for the additional force term due to buoyancy, the force vector is now given as

$$\begin{aligned}\bar{F} &= \bar{F}_{Gravity} + \bar{F}_{Buoyancy} + \bar{F}_{Aerodynamic} + \bar{F}_{Propulsive} \\ &= \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix}_E + \begin{bmatrix} 0 \\ 0 \\ -B \end{bmatrix}_E + \begin{bmatrix} X_{A/d} \\ Y_{A/d} \\ Z_{A/d} \end{bmatrix}_B + \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}_B\end{aligned}\quad (9.13)$$

Resultant components of gravity and buoyancy forces (to be used in stability analysis later) along body-fixed are

$$\bar{F}_{G+B}^B = \underbrace{\begin{bmatrix} -mg\sin\theta \\ mg\sin\phi\cos\theta \\ mg\cos\phi\cos\theta \end{bmatrix}}_{\bar{F}_{Gravity}^B} + \underbrace{\begin{bmatrix} B\sin\theta \\ -B\sin\phi\cos\theta \\ -B\cos\phi\cos\theta \end{bmatrix}}_{\bar{F}_{buoyancy}^B} = \begin{bmatrix} -(mg-B)\sin\theta \\ (mg-B)\sin\phi\cos\theta \\ (mg-B)\cos\phi\cos\theta \end{bmatrix}\quad (9.14)$$

Moment equations can be derived to be:

$$\bar{M} = m\bar{a} \times \left\{ \frac{d\bar{V}}{dt} \right|_B + \bar{\omega} \times \bar{V} \} + \left\{ \frac{d(\bar{I} \cdot \bar{\omega})}{dt} \right|_B + \bar{\omega} \times (\bar{I} \cdot \bar{\omega}) \} \quad (9.15)$$

Where $\bar{I}$ is inertia tensor. As should be obvious, the moments are also about the body fixed axes with contributions coming from various forces: aerodynamic, propulsive, buoyancy, weight, and other control elements (such as ballonets).

$$\begin{aligned}\bar{M} &= \bar{M}_{Aerodynamic} + \bar{M}_{prop} + \bar{M}_{gravity} + \bar{M}_{buyonacy} \\ &= \bar{r}_{A/d} \times \bar{F}_{A/d} + \bar{r}_p \times \bar{F}_{prop} + \bar{a} \times \bar{F}_{Gravity} + \bar{b} \times \bar{F}_{Buoyancy}\end{aligned}\quad (9.16)$$

Moments due to gravity and buoyancy forces are given by

$$\bar{M}_{G+B} = \begin{bmatrix} \mathcal{L} \\ M \\ N \end{bmatrix}_{G+B} = \bar{a} \times \bar{F}_{Gravity}^B + \bar{b} \times \bar{F}_{Buoyancy}^B$$

$$= \begin{bmatrix} \cos\phi\cos\theta(mga_y - Bb_y) + \sin\phi\cos\theta(Bb_z - mga_z) \\ \sin\theta(-mga_z + Bb_z) + \cos\phi\cos\theta(Bb_x - mga_x) \\ \sin\phi\cos\theta(-Bb_x + mga_x) + \sin\theta(-mga_y + Bb_y) \end{bmatrix} \quad (9.17)$$

One may arrive at complex arrangements of different control surfaces and their contributions to overall forces and moments can be determined using Eqs. 9.13 and 9.16. XZ-plane is the plane of symmetry for the airship, thus, $I_{xy} = I_{yx} = I_{zy} = I_{yz} = 0$, and CG lies in the plane of symmetry, that is, $a_y = 0$.

Components of $\bar{F}_{Aerodynamic}$ are defined as:

$$X_{A/d} = \frac{1}{2}\rho V^2 C_X Vol^{2/3}; Y_{A/d} = \frac{1}{2}\rho V^2 C_Y Vol^{2/3}; Z_{A/d} = \frac{1}{2}\rho V^2 C_Z Vol^{2/3} \text{ and}$$

$\mathcal{L}_{A/d} = \frac{1}{2}\rho V^2 C_l Vol; M_{A/d} = \frac{1}{2}\rho V^2 C_m Vol; N_{A/d} = \frac{1}{2}\rho V^2 C_n Vol$ are components aerodynamic moment $\bar{M}_{Aerodynamic}$.

Note that, in the above, volume of the airship has been used as reference for computing the coefficients unlike wing reference area and mean aerodynamic chord (as in the case of airplane) for arriving at non-dimensionalized aerodynamic force and moment coefficients.

A dominant feature that must be added to airship dynamic models are added/virtual mass and inertia effects. These effects are there for any body submerged in a medium, for example, our own body displaces air around us and we too do feel buoyancy (significantly higher in water) in general. But, as compared to our weight, this force is much smaller, actually negligible, and hence can be ignored. For airplanes, which actually displace huge amount of air around them due to their enormous outer volume, the force of buoyancy is negligible as compared to their enormous weight and lift produced due to relative air speed.

Airships are slow moving vehicles which may not produce large aerodynamic forces, but the unsteady aerodynamic effects due to their large size and being lighter-than-air are predominant. Apparent mass and inertia resulting from the unsteady aerodynamic effects are usually accounted for while modeling dynamics of an airship. These terms having dimension of mass/inertia are added to actual mass/inertia of the airship resulting in virtual/added mass/inertia.

Expressions for these terms are:

$$m_x = m - \frac{\partial X}{\partial \dot{u}}; \quad m_y = m - \frac{\partial Y}{\partial \dot{v}}; \quad m_z = m - \frac{\partial Z}{\partial \dot{w}} \quad (9.18)$$

Notice that the additional terms due to derivatives of aerodynamic contributions of X, Y, Z with respect to acceleration terms $\dot{u}, \dot{v}, \dot{w}$ have the unit of mass, thereby resulting in mass like terms.

Similarly, inertia terms are modified as:

$$J_{xx} = I_{xx} - \frac{\partial \mathcal{L}}{\partial \dot{p}}; J_{yy} = I_{yy} - \frac{\partial M}{\partial \dot{q}}; J_{zz} = I_{zz} - \frac{\partial N}{\partial \dot{r}} \quad (9.19)$$

In order to clearly see where added mass and inertia terms appear in the equations of motion, consider one example each from the force as well as moment equations. The X-axis force equation is (following Eq. (9.12))

$$\begin{aligned} \bar{F} &= m \left. \frac{d\bar{V}}{dt} \right|_B + m(\bar{\omega} \times \bar{V}) + m \left. \frac{d(\bar{\omega} \times \bar{a})}{dt} \right|_B + m\{\bar{\omega} \times (\bar{\omega} \times \bar{a})\} \\ m(\dot{u} + qw - rv) + ma_z \dot{q} - ma_y \dot{r} - ma_x(r^2 + q^2) + ma_z pr + ma_y pq \\ &= X_{A/d} - (mg - B)\sin\theta + T_R \cos\epsilon_R + T_L \cos\epsilon_L \end{aligned} \quad (9.20)$$

In the above equation, $T_R, T_L, \epsilon_R, \epsilon_L$ are thrust produced by the right and left engines and corresponding thrust vectoring angles. $X_{A/d}$ is the resultant aerodynamic force along body x -axis, which is function of states and rate of change of states as

$$X_{A/d} = X_{A/d}(u, v, w, p, q, r, \dot{u}, \dot{v}, \dot{w}, \dot{p}, \dot{q}, \dot{r}, \delta) \quad (9.22)$$

Where δ is aerodynamic control surface deflection. X can be separated as sum of a term ‘T1’ which is left out from the expansion and a term ‘T2’ which consists of rate derivative terms as,

$$X = \underbrace{X_R}_{T1} + \underbrace{X_{\dot{u}}\dot{u} + X_{\dot{q}}\dot{q} + X_{\dot{r}}\dot{r}}_{T2} \quad (9.23)$$

Where $X_{(\cdot)} = \frac{\partial X}{\partial (\cdot)}$.

Subscript ‘R’ has been used to represent the remaining terms. Term ‘T2’ when taken to the left hand side of Eq. 9.21 and setting $a_y = 0$, results in

$$(m - X_{\dot{u}})\dot{u} + mqw - mrv + (ma_z - X_{\dot{q}})\dot{q} - X_{\dot{r}}\dot{r} - ma_x(r^2 + q^2) + ma_z pr = X_R$$

Which is

$$m_x \dot{u} + mqw - mrv + (ma_z - X_{\dot{q}})\dot{q} - X_{\dot{r}}\dot{r} - ma_x(r^2 + q^2) + ma_z pr = X_R \quad (9.24)$$

Force equations along body fixed Y and Z axes are:

$$m_y \dot{v} + mru - mpw + (ma_x - Y_{\dot{r}})\dot{r} - (ma_z + Y_{\dot{p}})\dot{p} + ma_x pq + ma_z qr = Y_R \quad (9.25)$$

$$m_z \dot{w} + mpv - mqu - Z_{\dot{p}}\dot{p} - (ma_x + Z_{\dot{q}})\dot{q} - ma_z(p^2 + q^2) + ma_x pr = Z_R \quad (9.26)$$

Similarly, from expansion of Eq. 9.15

$$\bar{M} = m\bar{a} \times \left\{ \left. \frac{d\bar{V}}{dt} \right|_B + \bar{\omega} \times \bar{V} \right\} + \left\{ \left. \frac{d(\bar{I} \cdot \bar{\omega})}{dt} \right|_B + \bar{\omega} \times (\bar{I} \cdot \bar{\omega}) \right\}$$

The moment equations about different body axes can be written as (after dropping the cross inertia terms $I_{xy} = I_{yx} = I_{zy} = I_{yz} = 0$ and $a_y = 0$):

$$-ma_z\{\dot{v} + (ur - pw)\} + I_{xx}\dot{p} - I_{xz}\dot{r} - I_{xz}pq + I_{zz}rq - rqI_{yy} = \mathcal{L}_R + \mathcal{L}_{(\dot{\cdot})}(\dot{\cdot})$$

Collecting the terms with common derivatives, the above equation can be re-written as,

$$-\dot{v}(ma_z + \mathcal{L}_{\dot{v}}) - ma_z(ur - pw) + \underbrace{(I_{xx} - \mathcal{L}_{\dot{p}})}_{J_{xx}}\dot{p} - \underbrace{(I_{xz} + \mathcal{L}_{\dot{r}})}_{J_{xz}}\dot{r} - I_{xz}pq + I_{zz}rq - rqI_{yy} = \mathcal{L}_R \quad (9.27)$$

Similarly, other two moment equations including added inertia effect can be written as

$$(ma_z + M_{\dot{u}})\dot{u} + ma_z(qw - rv) - (ma_x + M_{\dot{w}})\dot{w} - ma_x(pv - uq) + \underbrace{(I_{yy} - M_{\dot{q}})}_{J_{yy}}\dot{q} + I_{xx}pr - I_{xz}(p^2 - r^2) - I_{zz}pr = M_R \quad (9.28)$$

$$(ma_x - N_{\dot{v}})\dot{v} + ma_x(ur - pw) - \underbrace{(I_{zx} + N_{\dot{p}})}_{J_{zx}}\dot{p} + \underbrace{(I_{zz} - N_{\dot{r}})}_{J_{zz}}\dot{r} - I_{xx}pq + I_{xz}rq + pqI_{yy} = N_R \quad (9.29)$$

Kinematic equations are (From Eq. 8.3, Chapter 8):

$$\dot{\phi} = p + \tan\theta(q\sin\phi + r\cos\phi)$$

$$\dot{\theta} = q\cos\phi - r\sin\phi$$

Trim condition: Our equilibrium condition of interest is a hover cruise trim state, thus $v^* = w^* = p^* = q^* = r^* = \phi^* = 0$; $u^* = \text{const} \neq 0$. Also assume, $\theta^* = 0$. Equilibrium condition requires that

$$X = -(mg - B)\sin\theta + T - D = 0$$

Which is, $T = D$ in both the cases, either $\theta^* = 0$ or $(mg - B) = 0$.

$$Y = 0$$

$$Z = (mg - B)\cos\theta - L = 0$$

Which is, $mg - B = L$ at $\theta^* = 0$. Trim angle of attack is zero and, therefore, aerodynamic lift produced on the airship is taken as zero at hover cruise condition satisfying $mg - B = 0$. Further, moment balance requires that

$$\mathcal{L} = M = N = 0.$$

Introducing perturbation in the state variables and control inputs, i.e.

$$\begin{aligned} u &= u^* + \Delta u; v = v^* + \Delta v; w = w^* + \Delta w; p = p^* + \Delta p; q = q^* + \Delta q; r = r^* + \Delta r; \phi \\ &= \phi^* + \Delta \phi; \theta = \theta^* + \Delta \theta; \delta(\cdot) = \delta(\cdot)^* + \Delta \delta(\cdot) \end{aligned}$$

in Eqs. 9.24-9.26 and Eqs. 9.27-9.29 and substituting values of equilibrium conditions, one could arrive at the small perturbation equations.

9.2.2 Longitudinal Small Perturbation Equations

For an example, let us try linearizing equation for forward motion (Eq. 9.24) at the chosen longitudinal flight condition. Substituting perturbed variables in Eq. 9.24, one may arrive at (dropping higher order and product of small Δ terms)

$$m_x \Delta \dot{u} + (ma_z - X_{\dot{q}}) \Delta \dot{q} = \Delta X_R$$

Which is,

$$m_x \Delta \dot{u} + (ma_z - X_{\dot{q}}) \Delta \dot{q} = \frac{\partial X}{\partial u} \Delta u + \frac{\partial X}{\partial w} \Delta w + \frac{\partial X}{\partial q} \Delta q + \frac{\partial X}{\partial \theta} \Delta \theta + \frac{\partial X}{\partial \delta} \Delta \delta \quad (9.30)$$

where, $\frac{\partial X}{\partial \theta} = -(mg - B)\cos\theta$. Similarly, the other longitudinal perturbation dynamics equations can be derived to be

$$m_z \Delta \dot{w} - mu^* \Delta q - (ma_x + Z_{\dot{q}}) \Delta \dot{q} = \Delta Z_R$$

$$m_z \Delta \dot{w} - mu^* \Delta q - (ma_x + Z_{\dot{q}}) \Delta \dot{q} = \frac{\partial Z}{\partial u} \Delta u + \frac{\partial Z}{\partial w} \Delta w + \frac{\partial Z}{\partial q} \Delta q + \frac{\partial Z}{\partial \theta} \Delta \theta + \frac{\partial Z}{\partial \delta} \Delta \delta \quad (9.31)$$

Where $\frac{\partial Z}{\partial \theta} = -(mg - B)\sin\theta$.

$$(ma_z + M_{\ddot{u}})\Delta\dot{u} - (ma_x + M_{\ddot{w}})\Delta\dot{w} + ma_x u^* \Delta q + \underbrace{(I_{yy} - M_{\dot{q}})}_{J_{yy}} \dot{\Delta q} = \Delta M_R$$

$$\begin{aligned} & (ma_z + M_{\ddot{u}})\Delta\dot{u} - (ma_x + M_{\ddot{w}})\Delta\dot{w} + ma_x u^* \Delta q + \underbrace{(I_{yy} - M_{\dot{q}})}_{J_{yy}} \dot{\Delta q} \\ &= \frac{\partial M}{\partial u} \Delta u + \frac{\partial M}{\partial w} \Delta w + \frac{\partial M}{\partial q} \Delta q + \frac{\partial M}{\partial \theta} \Delta \theta + \frac{\partial M}{\partial \delta} \Delta \delta \end{aligned} \quad (9.32)$$

Where $\frac{\partial M}{\partial \theta} = (-mga_z + Bb_z)\cos\theta + (mga_x - Bb_x)\sin\theta$ (From Eq. 9.17 with $\phi = 0$).

$$\Delta\dot{\theta} = \Delta q \quad (9.33)$$

Re-arranging and collecting the four linear equations for longitudinal dynamics, one would arrive at matrix form as

$$[M]\Delta\dot{x} = A\Delta x + B\Delta\delta \quad (9.34)$$

All Δ quantities are perturbation from equilibrium flight condition under study. $[M]$ is the mass matrix, which is,

$$M = \begin{bmatrix} m_x & 0 & (ma_z - X_{\dot{q}}) & 0 \\ 0 & m_z & -(ma_x + Z_{\dot{q}}) & 0 \\ (ma_z + M_{\ddot{u}}) & -(ma_x + M_{\ddot{w}}) & J_{yy} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9.35)$$

$$A = \begin{bmatrix} X_u & X_w & X_q & -(mg - B)\cos\theta^* \\ Z_u & Z_w & (Z_q + mu^*) & -(mg - B)\sin\theta^* \\ M_u & M_w & (M_q - ma_x u^*) & (-mga_z + Bb_z)\cos\theta^* + (mga_x - Bb_x)\sin\theta^* \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (9.36)$$

And matrix B, known as control matrix, is

$$B = \begin{bmatrix} X_{\delta} \\ Z_{\delta} \\ M_{\delta} \\ 0 \end{bmatrix} \quad (9.37)$$

For dynamic mode analysis, it suffices to set control input to zero and study the dynamics of the perturbation alone. Assuming general form of solution for $\Delta x(t) = e^{\lambda t} \Delta x(0)$ and substituting in Eq. 9.34 with $\delta = 0$ results in

$$(M\lambda - AI)\Delta x(t) = 0 \quad (9.38)$$

Which for $\Delta x(t) \neq 0$ gives the characteristic equation, $Det(M\lambda - AI) = 0$. As we know from before, location of eigenvalues will determine the stability of the equilibrium state of airship under consideration. This is the numerical procedure to determine stability of equilibrium/trim states. Assuming a level trim condition at which $w^* = 0$ (which also means angle of attack, $\alpha^* = 0$) and $\theta^* = 0$ (thus $\gamma^* = 0$), and also hover cruise condition at which $(mg - B) = 0$, one can arrive at further simplified form of the linear longitudinal dynamics equations.

Longitudinal dynamic modes: These are the modes characterized by dynamics of longitudinal perturbation variables $[\Delta u \ \Delta w \ \Delta q \ \Delta \theta]^T$ from the hover cruise condition as described earlier. Let us first look at decoupled form of equations for each mode, assuming the motions were constrained. Let us assume that airship is constrained to have pure forward or heave or pitch motion (3 degrees of freedom in longitudinal dynamics).

1. **Surge mode:** This is a first order mode representing motion primarily in forward velocity Δu . Following the perturbed equation,

$$m_x \Delta \dot{u} - \frac{\partial X}{\partial u} \Delta u = \frac{\partial X}{\partial \delta} \Delta \delta \quad (9.39)$$

$$\lambda_{surge} = \frac{X_u}{m_x} = \frac{X_u}{m - X_{\dot{u}}} \quad (9.40)$$

With the denominator positive always as $X_{\dot{u}}$ is significantly lower than the mass m , this mode is stable when the derivative X_u is negative. Expression for X_u , at equilibrium state other than hover, can be derived from

$$X = L \sin \alpha - D \cos \alpha + T - (mg - B) \sin \theta \quad (9.41)$$

$$\frac{\partial X}{\partial u} = \frac{\partial L}{\partial u} \sin \alpha - \frac{\partial D}{\partial u} \cos \alpha + \frac{\partial T}{\partial u} \quad (9.42)$$

Equilibrium state, $(mg - B) = 0$, has been used in the above and further using $w^* = 0$, the above expression simplifies to

$$\frac{\partial X}{\partial u} = -\frac{\partial D}{\partial u} + \frac{\partial T}{\partial u} = -u^* \rho_{air} S C_D^* + \frac{1}{2} \rho_{air} u^{*2} S (C_{T_u} - C_{D_u}) \quad (9.43)$$

Where $S = Vol^{2/3}$. The second term is significantly small at speeds at which airships operate. The first term is clearly negative indicating that the surge mode is stable. This is the slowest mode of the three. At the hover cruise condition, due to the drag dependency, variation of axial force with speed is however small, yet negative, hence stabilizing.

2. **Heave mode:** This is again a first order mode representing motion primarily in vertical velocity Δw governed by

$$m_z \Delta \dot{w} - \frac{\partial Z}{\partial w} \Delta w = \frac{\partial Z}{\partial \delta} \Delta \delta \quad (9.44)$$

Thus, approximate expression for heave mode eigenvalue is

$$\lambda_{heave} = \frac{Z_w}{m_z} \quad (9.45)$$

$$Z = (mg - B)\cos\theta - L\cos\alpha - D\sin\alpha \quad (9.46)$$

From which expression for Z_w can be derived as

$$Z_w = \frac{\partial Z}{\partial w} = \frac{1}{u^*} \frac{\partial Z}{\partial \alpha} = \frac{1}{u^*} (L\sin\alpha - L_\alpha \cos\alpha - D\cos\alpha - D_\alpha \sin\alpha) \quad (9.47)$$

At equilibrium state $\alpha^* = 0$, the above expression is reduced to

$$Z_w = \frac{1}{u^*} (-L_\alpha - D) \quad (9.48)$$

Drag is much larger as compared to L_α , which is also positive, thus contributing to stability of the heave mode. With increasing trim speed, however, these above approximate expressions can be modified by including more coupling effects.

3. **Pitch oscillatory mode:** This is a second order mode representing oscillatory motion in pitch variable, $\Delta\theta$ or $\Delta\alpha$. Usually, heave and the pitch oscillatory modes are strongly coupled.

In order to get some realistic feel for this mode, we consider pure pitching motion of airship governed by

$$\begin{aligned} (ma_z + M_{\ddot{u}})\Delta \ddot{u} - (ma_x + M_{\ddot{w}})\Delta \ddot{w} + ma_x u^* \Delta q + \underbrace{(I_{yy} - M_{\dot{q}})}_{J_{yy}} \Delta \dot{q} \\ = \frac{\partial M}{\partial u} \Delta u + \frac{\partial M}{\partial w} \Delta w + \frac{\partial M}{\partial q_1} \Delta q + \frac{\partial M}{\partial \theta} \Delta \theta + \frac{\partial M}{\partial \delta} \Delta \delta \end{aligned} \quad (9.49)$$

Since $\Delta \dot{\alpha} = \Delta \dot{\theta} = \Delta q$,

$$J_{yy}\Delta\dot{q} = \frac{\partial M}{\partial w}\Delta w + \frac{\partial M}{\partial \dot{w}}\Delta\dot{w} + \left(\frac{\partial M}{\partial q_1} - ma_x u^*\right)\Delta\dot{\alpha} + \frac{\partial M}{\partial \theta}\Delta\theta + \frac{\partial M}{\partial \delta}\Delta\delta \quad (9.50)$$

Which (with $\Delta\alpha \approx \frac{\Delta w}{u^*}$) can be re-written as

$$J_{yy}\Delta\dot{q} = \frac{\partial M}{\partial \alpha}\Delta\alpha + \frac{\partial M}{\partial \dot{\alpha}}\Delta\dot{\alpha} + \left(\frac{\partial M}{\partial q_1} - ma_x u^*\right)\Delta\dot{\alpha} + \frac{\partial M}{\partial \theta}\Delta\theta + \frac{\partial M}{\partial \delta}\Delta\delta \quad (9.51)$$

Which is,

$$J_{yy}\Delta\ddot{\alpha} = M_\alpha\Delta\alpha + (M_{\dot{\alpha}} + M_{q_1} - ma_x u^*)\Delta\dot{\alpha} + M_\theta\Delta\alpha + M_\delta\Delta\delta \quad (9.52)$$

Where $M_\theta = \left.\frac{\partial M}{\partial \theta}\right|_* = (-mga_z + Bb_z)$

Re-arranging Eq. 9.52, we get

$$J_{yy}\Delta\ddot{\alpha} - (-ma_x u^* + M_{q_1} + M_{\dot{\alpha}})\Delta\dot{\alpha} - (M_\alpha - mga_z + Bb_z)\Delta\alpha = M_\delta\Delta\delta \quad (9.53)$$

A quick inspection of Eq. 9.53 tells us that it is similar to Eq. 2.32 except few additional terms. For stability of airship at the hover condition ($u^* = 0$)

$$(M_{q_1} + M_{\dot{\alpha}}) < 0 \text{ and } (M_\alpha - mga_z + Bb_z) < 0 \quad (9.54)$$

Popularly used Munk's relation for aerodynamic pitching moment often used in the context of fuselage (and similar looking hull shape of airship)

$$M = \frac{1}{2}\rho V^2(Vol)(k_2 - k_1)\sin 2\alpha \quad (9.55)$$

gives, upon differentiation with respect to the angle of incidence, also angle of attack,

$$M_\alpha = \rho V^2(Vol)(k_2 - k_1)\cos 2\alpha \quad (9.56)$$

The proportionality factor, $(k_2 - k_1)$, related to Lamb's inertia ratios, is always positive meaning airship hull contribution to pitch static stability is destabilizing. The CG is always aft of the lift point for airship with hull of elliptic cylindrical shape. From the above relations, expression for lift produced (known to be significantly small) and lift curve slope can be estimated. Similar to airplanes, aft tails (fins) are designed to contribute to stability of airship. Other contributions to positive stiffness and damping result from purely geometric considerations:

- If CG is placed slightly aft of the reference Z-axis in the longitudinal plane, it may result in positive damping (but the contributions from stabilator depending on the moment arm length may reduce in that case), and

- $a_z > b_z$ for positive stiffness in pitch (remember $(mg - B) = 0$ at our equilibrium condition).

What it means is, CG must lie below the point at which force of buoyancy acts. This is in general true as most of the payload is concentrated near bottom of the hull.

From Eq. 9.53, one may obtain approximate expressions for pitch pendulum mode frequency and damping ratio from

$$\omega_{np}^2 = -\frac{(M_{\alpha} - mga_z + Bb_z)}{J_{yy}}; \quad 2\zeta_p \omega_{np} = -\frac{(ma_x u^* + M_{q1} + M_{\dot{\alpha}})}{J_{yy}}. \quad (9.57)$$

9.2.3 Small perturbation equations for lateral-directional modes

$$m_y \Delta \dot{v} + (ma_x - Y_{\dot{r}}) \Delta \dot{r} - (ma_z + Y_{\dot{p}}) \Delta \dot{p} = \frac{\partial Y}{\partial v} \Delta v + \frac{\partial Y}{\partial p} \Delta p + \left(\frac{\partial Y}{\partial r} - mu^* \right) \Delta r + \frac{\partial Y}{\partial \phi} \Delta \phi + \frac{\partial Y}{\partial \delta} \Delta \delta \quad (9.58)$$

$$(ma_x - N_{\dot{v}}) \Delta \dot{v} - J_{xz} \Delta \dot{p} + J_{zz} \Delta \dot{r} = \frac{\partial N}{\partial v} \Delta v + \frac{\partial N}{\partial p} \Delta p + \left(\frac{\partial N}{\partial r} - ma_x u^* \right) \Delta r + \frac{\partial N}{\partial \phi} \Delta \phi + \frac{\partial N}{\partial \delta} \Delta \delta \quad (9.59)$$

$$-\Delta \dot{v} (ma_z + L_{\dot{v}}) + J_{xx} \Delta \dot{p} - J_{xz} \Delta \dot{r} = \frac{\partial L}{\partial v} \Delta v + \frac{\partial L}{\partial p} \Delta p + \left(\frac{\partial L}{\partial r} + ma_z u^* \right) \Delta r + \frac{\partial L}{\partial \phi} \Delta \phi + \frac{\partial L}{\partial \delta} \Delta \delta \quad (9.60)$$

$$\Delta \dot{\phi} = \Delta p \quad (9.61)$$

These equations can be put in matrix form as (with $\Delta x = [\Delta v \ \Delta p \ \Delta r \ \Delta \phi]^T$)

$$M = \begin{bmatrix} m_y & -(ma_z - Y_{\dot{p}}) & (ma_x - Y_{\dot{r}}) & 0 \\ -(ma_z + L_{\dot{v}}) & J_{xx} & -J_{xz} & 0 \\ (ma_x - N_{\dot{v}}) & -J_{xz} & J_{zz} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9.62)$$

$$A = \begin{bmatrix} Y_v & Y_p & (Y_r - mu^*) & (mg - B) \cos \theta^* \\ L_v & L_p & (L_r + ma_z u^*) & (-mga_z + Bb_z) \cos \theta^* \\ N_v & N_p & (N_r - ma_x u^*) & (mga_x - Bb_x) \cos \theta^* \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (9.63)$$

Lateral-directional modes: These are the dynamic modes characterized by dynamics in remaining off-longitudinal perturbation variables $[\Delta v \ \Delta p \ \Delta r \ \Delta \phi]^T$. It is a formidable task to derive modal approximations in this case, however, some conclusions may be drawn from simplified analysis.

1. **Roll pendulum mode:** This mode is characterized by lightly damped oscillation in $\Delta\phi$. From pure rolling motion governed by

$$J_{xx}\Delta\dot{p} = \frac{\partial\mathcal{L}}{\partial p}\Delta p + \frac{\partial\mathcal{L}}{\partial\phi}\Delta\phi + \frac{\partial\mathcal{L}}{\partial\delta}\Delta\delta$$

$$\Delta\dot{\phi} = \Delta p$$

$$J_{xx}\Delta\ddot{\phi} - \frac{\partial\mathcal{L}}{\partial p}\Delta\dot{\phi} - \frac{\partial\mathcal{L}}{\partial\phi}\Delta\phi = \frac{\partial\mathcal{L}}{\partial\delta}\Delta\delta \quad (9.64)$$

One can see that the stiffness in this mode is provided by the derivative $\frac{\partial\mathcal{L}}{\partial\phi}$, which at the trim condition $\phi^* = \theta^* = 0$ is $\frac{\partial\mathcal{L}}{\partial\phi} = (Bb_z - mga_z)$. Further, at hover condition $mg = B$, hence $\frac{\partial\mathcal{L}}{\partial\phi} = mg(b_z - a_z)$. For positive stiffness in roll $\frac{\partial\mathcal{L}}{\partial\phi}$ must be negative, therefore, $b_z < a_z$, a condition which was also required for positive stiffness in pitch.

2. **Yaw subsidence mode:** Characterized by first order fast dynamics in Δr , and

3. **Sideslip subsidence mode:** Characterized by first order dynamics in Δv or $\Delta\beta$. Yaw subsidence mode is the fastest among the three, while roll oscillation and sideslip subsidence modes have a different slower time scale of dynamics.

9.2.4 Useful empirical relations

Some estimates for the virtual mass and inertia terms can be obtained from the relations

$$X_{\ddot{u}} = -k_1 \frac{B}{g}; \quad Y_{\ddot{v}} = -k_2 \frac{B}{g}; \quad Z_{\ddot{w}} = Y_{\ddot{v}}; \quad M_{\ddot{q}} = -\frac{k' \left(\frac{B}{g}\right) (l^2 + d^2)}{20}; \quad N_{\ddot{r}} = M_{\ddot{q}}$$

where l is the length and d is the maximum diameter of the hull. The factors k_1, k_2, k' known as Lamb's inertia ratios are function of $\left(\frac{l}{d}\right)$ or the fineness ratio.

9.2.5 Numerical example

Data for a Gertler² shape airship operating at stratospheric altitude of 20km ($\rho_{air} = 0.0767 \text{ kg/m}^3$) at $\alpha^* = 0, u^* = 15 \text{ m/s}$ are given as:

$$m = 25093.172kg; m_x = 27602.492kg; m_y = 43044.16kg; m_z = 43044.16kg;$$

$$J_{xx} = 17400000.0kg - m^2; J_{yy} = 245264282.0kg - m^2;$$

$$J_{zz} = J_{yy}; J_{xz} = 1920000.0kg - m^2$$

$$Vol = 327160m^3; l = 226m; d = 56.5m; l/d = 4;$$

$$b_z = 16.43m; a_z = 18.0m; a_x = 0; b_x = 0;$$

$$k_1 = 0.1; k_2 = 0.9; k' = 0.6$$

$$B = Vol\rho_{air}g; B/g = 327160 \times 0.0767 = 25093.172kg$$

$$X_{\dot{u}} = -k_1 \frac{B}{g} = -2509.32; Y_{\dot{v}} = -k_2 \frac{B}{g} = -22583.85; Z_{\dot{w}} = Y_{\dot{v}}$$

$$M_{\dot{q}} = -k' \frac{B}{g} \times \frac{(l^2 + d^2)}{20} = -0.6 \times 25093.17 \times \frac{(226^2 + 56.5^2)}{20} = -40852875.94$$

$$N_{\dot{r}} = M_{\dot{q}};$$

$$M_{\alpha} = \frac{1}{2} \rho_{air} u^{*2} Vol (k_2 - k_1) 2 = 0.0767 \times 15^2 \times 327160 \times (0.9 - 0.1) = 4516770.96;$$

$$M_w = \frac{M_{\alpha}}{u^*} = 301118.0;$$

$$M_q = \frac{1}{2} \rho u^{*2} Vol C_{mq1} = 0.5 \times 0.0767 \times 15^2 \times 327160 \times -0.065 = -183493.82$$

$$Z_q = -\frac{1}{2} \rho u^{*2} Vol l^{\frac{2}{3}} C_{Lq1} = -0.5 \times 0.0767 \times 15^2 \times 327160^{\frac{2}{3}} \times 0.021 = -860.35$$

$$X_u = -\rho u^{*2} Vol l^{\frac{2}{3}} C_D^* = -0.0767 \times 15 \times 327160^{\frac{2}{3}} \times 0.03 = -163.88$$

$$Z_w = -\frac{1}{2} \rho u^{*2} Vol l^{\frac{2}{3}} C_D^* = \frac{1}{2} X_u = -81.94$$

$$M_{\theta} = (-mga_z + Bb_z) = mg(b_z - a_z) = -386477.51$$

$$Y_{\beta} = \frac{1}{2} \rho u^{*2} Vol l^{2/3} C_{y\beta} = 0.5 \times 0.0767 \times 15^2 \times 327160^{2/3} \times -0.0109 = -446.56$$

$$Y_v = \frac{Y_\beta}{u^*} = -\frac{446.561}{15} = -29.77; Y_p = 0$$

$$Y_r = \frac{1}{2}\rho u^{*2} Vol^{2/3} C_{yr} = 0.5 \times 0.0767 \times 15^2 \times 327160^{2/3} \times 0.021 = 860.35; Y_\varphi = 0$$

$$\mathcal{L}_\beta = \frac{1}{2}\rho u^{*2} Vol C_{l\beta} = \frac{1}{2}\rho u^{*2} Vol \times 0.0 = 0.0$$

$$\mathcal{L}_p = \frac{1}{2}\rho u^{*2} Vol C_{lp} = 0.5 \times 0.0767 \times 15^2 \times 327160 \times -0.002 = -5645.96$$

$$\mathcal{L}_r = \frac{1}{2}\rho u^{*2} Vol C_{lr} = 0.5 \times 0.0767 \times 15^2 \times 327160 \times 0.0 = 0.0$$

$$\mathcal{L}_\varphi = mg(b_z - a_z) = -386477.51$$

$$N_\beta = \frac{1}{2}\rho u^{*2} Vol C_{n\beta} = 0.5 \times 0.0767 \times 15^2 \times 327160 \times -0.00251 = -7085.68$$

$$N_v = \frac{N_\beta}{u^*} = -472.38$$

$$N_\varphi = mg(a_x - b_x) = 0$$

Longitudinal Modes:

$$M_{long} = \begin{bmatrix} 25032.64 & 0 & 416610.0 & 0 \\ 0 & 43044.16 & 0 & 0 \\ 416610.0 & 0 & 245264282.0 & 0 \\ 0 & 0 & 0 & 1.0 \end{bmatrix}$$

$$A_{long} = \begin{bmatrix} -163.875 & 0 & 0 & 0 \\ 0 & -81.937 & 860.347 & 0 \\ 0 & 301118.0 & -183493.82 & -386477.51 \\ 0 & 0 & 1.0 & 0 \end{bmatrix}$$

Computed eigenvalues of matrix $M_{long}^{-1}A_{long}$:

$$\underbrace{-0.0005 \pm j0.04}_{pitch\ oscillatory}, \underbrace{-0.0066}_{surge}, \underbrace{-0.0019}_{heave}$$

Lateral-Directional Modes:

$$M = \begin{bmatrix} 43044.16 & -416610 & 0 & 0 \\ -416610 & 17400000 & -1920000 & 0 \\ 0 & -1920000 & 245264282 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{lat-dir} = \begin{bmatrix} -29.77 & 0 & 860.35 & 0 \\ 0 & -5645.96 & 0 & -386477.51 \\ -472.38 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Computed eigenvalues of $M_{lat-dir}^{-1}A_{lat-dir}$:

$$\underbrace{-0.0003 \pm j0.17}_{\text{Roll-pendulum}}, \quad \underbrace{-0.0006}_{\text{yaw subsidence}}, \quad \underbrace{-0.0001}_{\text{sideslip subsidence}}$$

References:

1. Khatri, A.K., Singh, J., and Sinha, N.K., Aircraft Design using Constrained Bifurcation and Continuation Method, *Journal of Aircraft*, Vol. 51, No. 5, 2014, pp. 1647-1652.
2. Rana, V., Development of Flight Dynamic Simulation Model of Stratospheric Airship, *Masters Dissertation*, Department of Aerospace Engineering, IIT Madras, 2014.